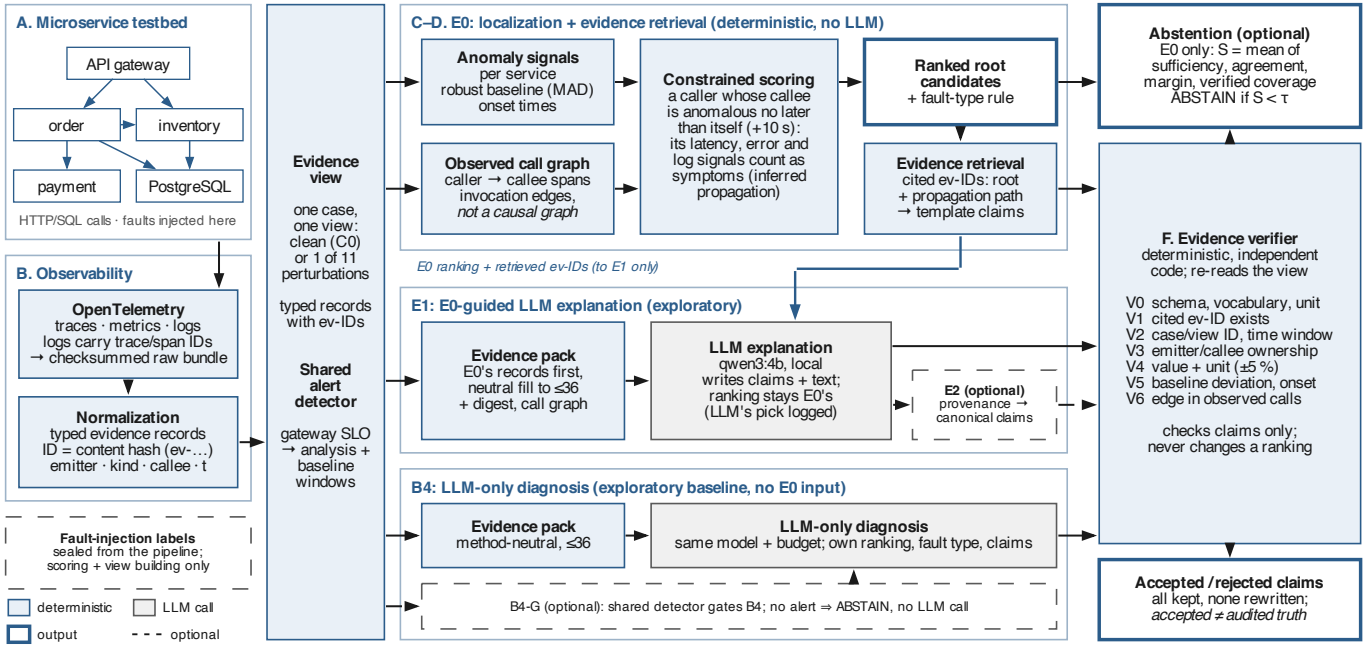

A. Microservice testbed
API gateway
order
inventory
payment
PostgreSQL
HTTP/SQL calls · faults injected here
B. Observability
OpenTelemetry
traces · metrics · logs
logs carry trace/span IDs
→ checksummed raw bundle
Normalization
typed evidence records
ID = content hash (ev-…)
emitter · kind · callee · t
Fault-injection labels
sealed from the pipeline;
scoring + view building only
deterministic
LLM call
output
optional
Evidence view
one case,
one view:
clean (C0)
or 1 of 11
perturbations
typed records
with ev-IDs
Shared alert detector
gateway SLO
→ analysis +
baseline
windows
C–D. E0: localization + evidence retrieval (deterministic, no LLM)
Anomaly signals
per service
robust baseline (MAD)
onset times
Observed call graph
caller → callee spans
invocation edges,
not a causal graph
Constrained scoring
a caller whose callee
is anomalous no later
than itself (+10 s):
its latency, error and
log signals count as
symptoms (inferred
propagation)
Ranked root candidates
+ fault-type rule
Evidence retrieval
cited ev-IDs: root
+ propagation path
→ template claims
E0 ranking + retrieved ev-IDs (to E1 only)
E1: E0-guided LLM explanation (exploratory)
Evidence pack
E0's records first,
neutral fill to ≤36
+ digest, call graph
LLM explanation
qwen3:4b, local
writes claims + text;
ranking stays E0's
(LLM's pick logged)
E2 (optional)
provenance →
canonical claims
B4: LLM-only diagnosis (exploratory baseline, no E0 input)
Evidence pack
method-neutral, ≤36
LLM-only diagnosis
same model + budget; own ranking, fault type, claims
B4-G (optional): shared detector gates B4; no alert ⇒ ABSTAIN, no LLM call
Abstention (optional)
E0 only: S = mean of
sufficiency, agreement,
margin, verified coverage
ABSTAIN if S < τ
F. Evidence verifier
deterministic, independent
code; re-reads the view
V0 schema, vocabulary, unit
V1 cited ev-ID exists
V2 case/view ID, time window
V3 emitter/callee ownership
V4 value + unit (±5 %)
V5 baseline deviation, onset
V6 edge in observed calls
checks claims only;
never changes a ranking
Accepted / rejected claims
all kept, none rewritten;
accepted ≠ audited truth